

Supplemental Data

다중장기 전사체 통합분석을 통한 *Angptl4*-*Syndecan*/*Cadherin* 축 매개 심장-간 증후군 신규 표적 발굴 및
구조 기반 약물 스크리닝 — Supplemental Figures and Tables

Supplemental Figures

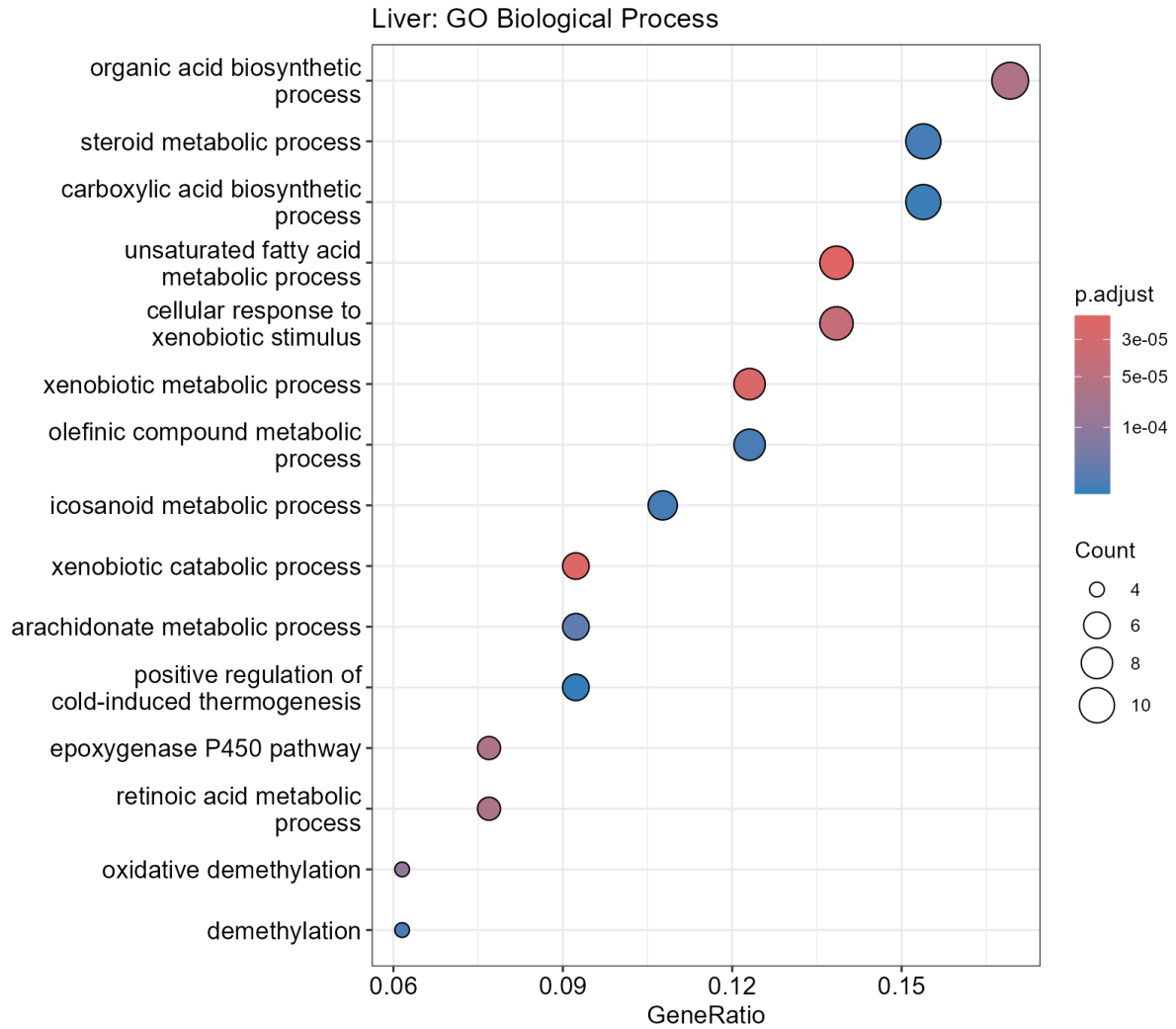


Figure S1. Liver GO Biological Process enrichment (34 significant terms, $p_{adj} < 0.05$).

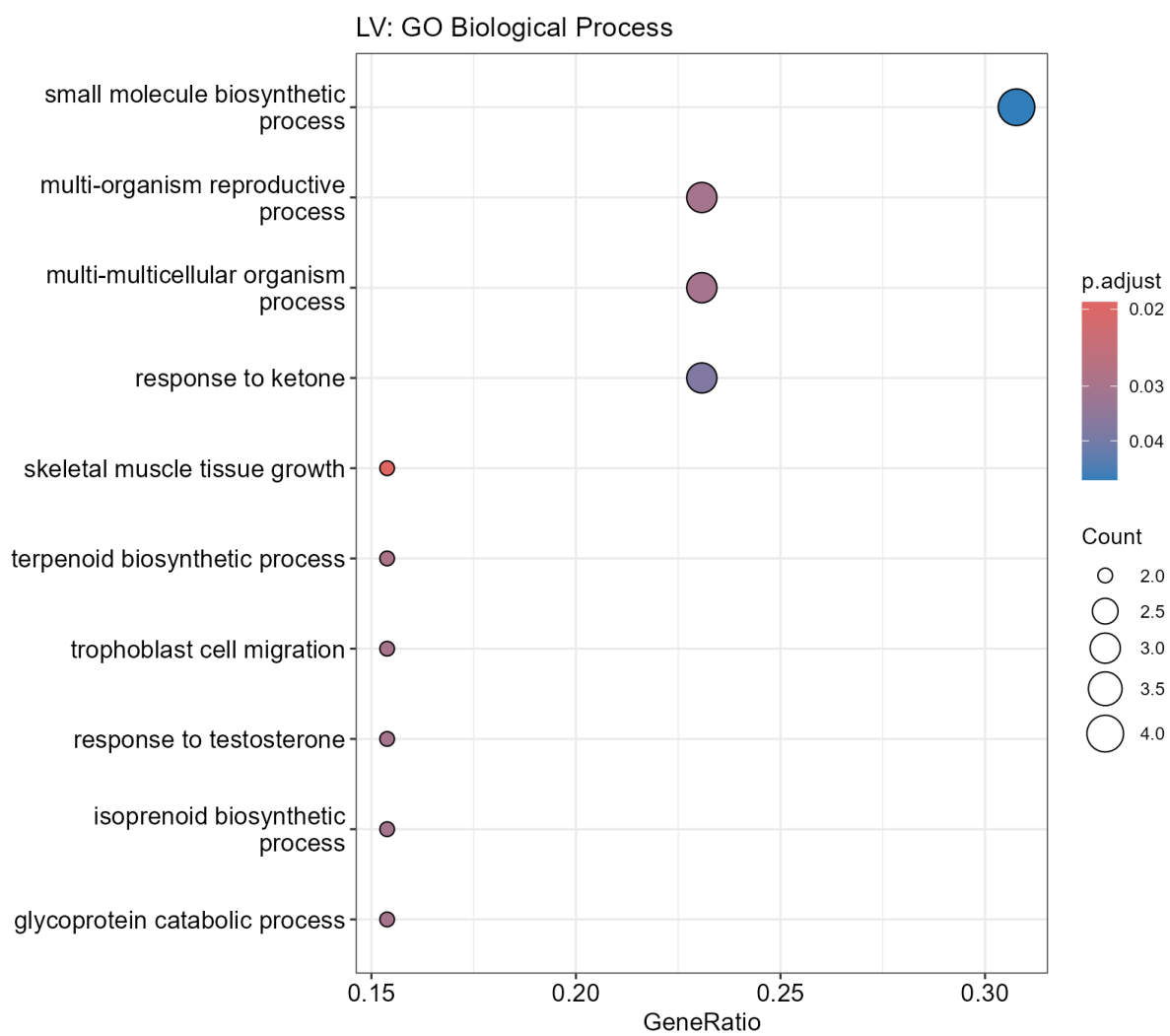


Figure S2. LV GO Biological Process enrichment (10 significant terms, $p_{adj} < 0.05$).

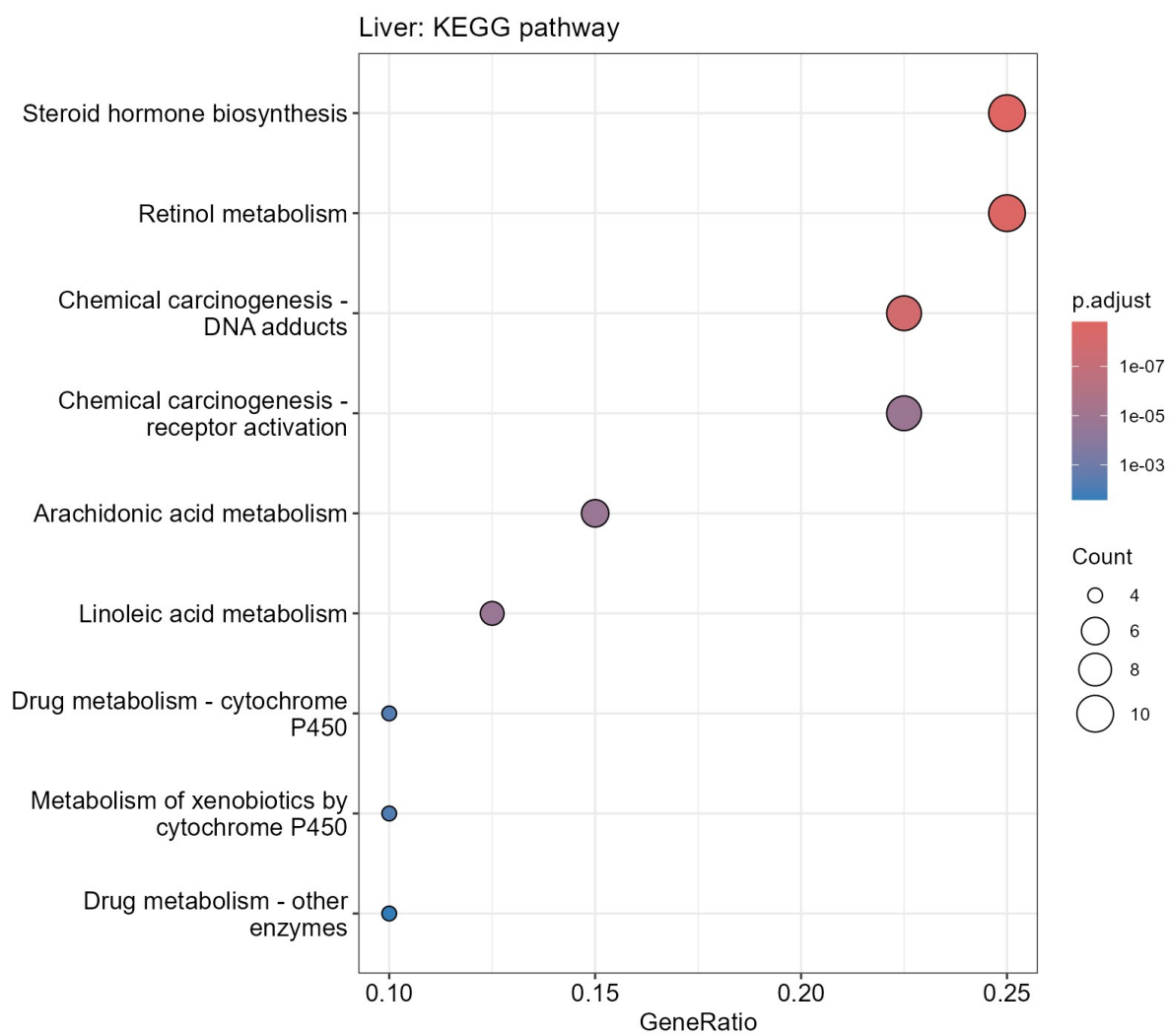


Figure S3. Liver KEGG pathway enrichment (9 significant pathways, $p_{adj} < 0.05$). No KEGG pathway reached significance in LV.

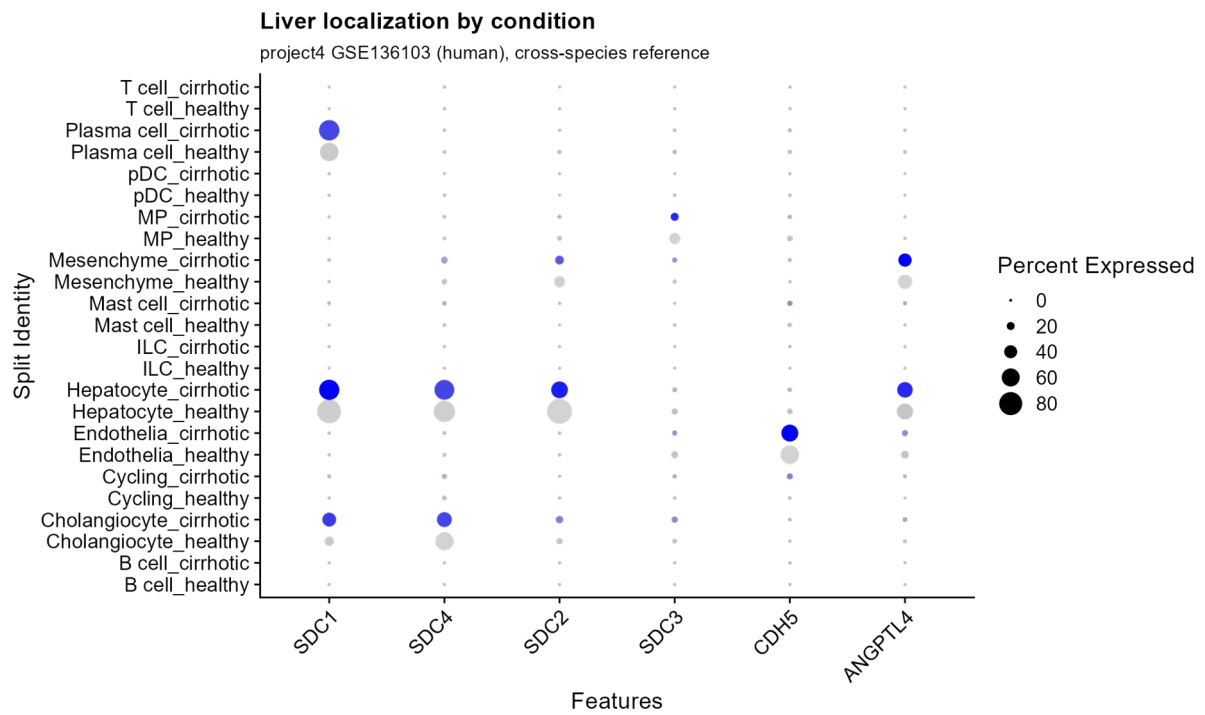


Figure S4. Liver localization of Angptl4-axis genes split by condition (healthy vs. cirrhotic; project4 GSE136103 cross-species reference).

Supplemental Tables

Table S1. Full significant DEG list, Liver (n=86, padj<0.01, |log2FC|>1.5)

ensembl_id	baseMean	log2FoldChange	lfcSE	pvalue	padj	gene_name
ENSMUSG00000034774	503.475388306876	2.64429327634708	0.212996384936588	1.39537690151581e-36	2.02469188409944e-32	Dsg1c
ENSMUSG00000040616	233.084960747301	-1.57213782341264	0.155920994817588	3.88811679866027e-25	1.41041436871401e-21	Tmem51
ENSMUSG00000025002	184.776235730195	-2.95549606645123	0.298917764494612	2.45979852017676e-24	5.94861275462746e-21	Cyp2c55
ENSMUSG00000102095	1213.4461453138	2.34989330530966	0.238929661089155	4.28063697314542e-24	8.87314892576287e-21	C730036E19Rik
ENSMUSG00000004038	1671.89161257983	-2.30069790272563	0.246866137221666	5.60733540057556e-22	1.01703045827939e-18	Gstm3
ENSMUSG00000097426	757.931557207992	2.28745881411989	0.254033251749858	1.17539446263632e-20	1.89499707253922e-17	Gm8941
ENSMUSG00000038060	62.4375612302593	2.41883223232349	0.285631383979538	1.20429163712742e-18	1.52184952092415e-15	Dlec1
ENSMUSG00000085834	232.693670444981	3.57622385332592	0.424846999216286	1.25859367684974e-18	1.52184952092415e-15	Gm15622
ENSMUSG00000040660	461.654325493277	4.76612271546659	0.576013184410581	5.53051412404958e-18	6.17290461076611e-15	Cyp2b9
ENSMUSG00000048368	86.633193258359	1.53140854822275	0.197566523167157	4.63189442300078e-16	4.65051132709902e-13	Omd
ENSMUSG00000055866	1283.92773618528	1.54300995671552	0.199506431618138	5.25542389511422e-16	4.76601254488171e-13	Per2
ENSMUSG00000030257	165.45506970847	-2.8895867942592	0.377010856772018	9.15700661810583e-16	7.81577447227738e-13	Srgap3
ENSMUSG00000026475	2092.78574519369	3.64609583478058	0.47804254562425	1.14747928863316e-15	9.24995804337065e-13	Rgs16
ENSMUSG00000005677	3000.74783767299	1.58700471125647	0.212480251653979	4.12265691679091e-15	3.14840799277032e-12	Nr1i3
ENSMUSG00000023044	6673.59478721416	-2.41062007521851	0.325742646236205	6.25723758426475e-15	4.53962586738408e-12	Csad
ENSMUSG00000069170	251.592851664664	2.16651042225976	0.293105763859143	8.06590679703661e-15	5.4017476922126e-12	Adgrv1
ENSMUSG00000021259	98.7214766767695	1.57370677646021	0.217545271349545	1.94822058822189e-14	1.09348856000731e-11	Cyp46a1
ENSMUSG00000099568	19.1449077349142	4.34751291066598	0.58927805915887	4.84230967169531e-14	2.50935404772496e-11	Gm28513
ENSMUSG00000045672	2462.59697503316	1.54499356864996	0.216470787518036	4.84099137795888e-14	2.50935404772496e-11	Col27a1
ENSMUSG00000030137	43.2872368975039	-2.58353717790484	0.36523637624022	6.48414079516269e-14	3.24430630820037e-11	Tuba8
ENSMUSG00000110496	146.126556323611	2.57568270325534	0.366567136943467	9.75018475899839e-14	4.71583936176889e-11	Gm45909
ENSMUSG00000108825	195.710461902802	1.55720788262107	0.221565285877381	1.15524818052276e-13	5.40730680625329e-11	Gm45838
ENSMUSG00000105391	49.3606698062927	1.98512987136949	0.299921203596909	1.44201892492306e-12	5.65505259476582e-10	Gm43401
ENSMUSG00000029822	237.08162061726	2.15505014321603	0.335962066861222	6.46341694714879e-12	2.28741902202753e-09	Osbpl3
ENSMUSG00000090175	516.646697485	-2.23470003682453	0.361619080246087	2.87808381998234e-11	9.07847744085733e-09	Ugt1a9
ENSMUSG00000057465	690.886491056676	1.78114533656839	0.29329685555328	5.61399049943249e-11	1.62918004293531e-08	Saa2
ENSMUSG00000085211	83.5853289326452	2.27299239156911	0.381437351389097	1.01331119158786e-10	2.74957833813579e-08	B430219N15Rik
ENSMUSG00000096215	81.7240445139723	-1.51458742919241	0.25397202093263	1.02327519131174e-10	2.74957833813579e-08	Smim22
ENSMUSG00000026822	143.919404952623	1.7747824470032	0.305858713103111	2.96510237092158e-10	7.17060590034535e-08	Lcn2

ensembl_id	baseMean	log2FoldChange	lfcSE	pvalue	padj	gene_name
ENSMUSG00000032482	105.343357814561	-2.14370854029406	0.374939490019592	5.29145372862328e-10	1.21871418416387e-07	Cspg5
ENSMUSG00000061906	407.471077450847	-1.96973865869983	0.348547255980042	6.99493400180646e-10	1.51487302039122e-07	Ugt2b38
ENSMUSG00000028957	1220.41286940673	1.59277260353284	0.287416193049593	1.39738912020119e-09	2.8161272408499e-07	Per3
ENSMUSG00000061292	1015.4361831682	-1.88983330776215	0.356498362621404	4.997147882676e-09	9.17830579463655e-07	Cyp3a59
ENSMUSG00000029254	51.497023631564	1.897757223332	0.364838568194038	8.4090970410215e-09	1.40248273638186e-06	Stap1
ENSMUSG00000022351	1927.2188631814	-1.64467428029207	0.324712848462326	1.81916911200905e-08	2.66121461743334e-06	Sqle
ENSMUSG000000108425	79.8323823453028	1.57831612368138	0.312015737650785	2.27885225064761e-08	3.14915677684731e-06	Gm44706
ENSMUSG00000032773	17.487534249692	2.25711338050943	0.453268229847169	2.58531317852317e-08	3.50587796452068e-06	Chrm1
ENSMUSG00000027533	1000.06428063829	-2.79306755072058	0.570708158333458	3.88111120324011e-08	5.1195385053649e-06	Fabp5
ENSMUSG00000030302	47.1906926893873	-1.75980397182386	0.358818599568666	4.18218637123124e-08	5.41817180772905e-06	Atp2b2
ENSMUSG00000049109	63.0119494150958	2.53186062101198	0.518540976460581	4.1553296293353e-08	5.41817180772905e-06	Themis
ENSMUSG00000086245	108.32248138857	1.58443107419618	0.324445748864558	5.35431450050601e-08	6.64026524806344e-06	Gm16170
ENSMUSG00000097762	81.651463977587	1.81925116292515	0.384014851633741	9.0015745869005e-08	1.07944501864402e-05	4732463B04Rik
ENSMUSG00000089712	100.580184766758	1.80356149676301	0.38842663020433	1.40830651123545e-07	1.59644745922081e-05	Gm15889
ENSMUSG00000001076	17.02603990295	-1.70405875993617	0.365141246779795	1.69231016012441e-07	1.86025912298524e-05	C1ql4
ENSMUSG00000049723	20.0381554698832	2.07998968568883	0.447192498406029	1.85165569632691e-07	2.00503911594802e-05	Mmp12
ENSMUSG00000057933	576.196153284774	-1.76891069316312	0.385887177021199	1.89327750344454e-07	2.03492270925779e-05	Gsta2
ENSMUSG00000012123	35.4531322336344	1.65101650446626	0.362947647808665	2.39274357106565e-07	2.47990780115447e-05	Crybg2
ENSMUSG00000040583	17.2718000888921	3.88076851176883	0.869928156891006	3.74043305763701e-07	3.54729958603353e-05	Cyp2b13
ENSMUSG00000090105	23.2866504862879	1.54441267413779	0.346895517391425	4.06125678891998e-07	3.77748948764288e-05	Gm15890
ENSMUSG00000092008	46.6255327924104	-2.64333323561804	0.599234862145356	4.46404324338968e-07	4.02318431438411e-05	Cyp2c69
ENSMUSG00000075044	21.1374986591566	2.13622927040766	0.490388947072474	5.6067350284095e-07	4.89463689651337e-05	Slc22a29
ENSMUSG00000026622	62.252521168354	-1.53132834673788	0.354434296556186	7.09378466082344e-07	5.93321979152194e-05	Nek2
ENSMUSG00000027068	82.6238537349016	-2.814860625705	0.665050568618433	8.51820731959965e-07	6.64511764555866e-05	Dhrs9
ENSMUSG00000056978	487.384642782162	1.58898036242832	0.374757650569082	9.75852495665254e-07	7.37480193338689e-05	Hamp2
ENSMUSG00000037071	175490.135891387	-2.02412764651439	0.487542574417716	1.26180457426227e-06	9.16514088912715e-05	Scd1
ENSMUSG00000067656	26.2965331234244	2.46144330277243	0.590769149946265	1.33466790904984e-06	9.4009860972394e-05	Slc22a27
ENSMUSG00000092035	22.3035997476215	-1.71998464310816	0.416284049827858	1.77080224543639e-06	0.00011786394762056	Peg10
ENSMUSG00000006398	24.3137100703418	-2.43262130844438	0.599393330092832	2.10459567280823e-06	0.000132772535706293	Cdc20
ENSMUSG00000095061	68.3373006070974	1.71155768038216	0.427769779770463	2.45326589560022e-06	0.000150197840274933	E030018B13Rik
ENSMUSG00000056035	65946.37167539	-2.70741979301712	0.688205414363059	2.86541008749761e-06	0.000166976306705182	Cyp3a11

ensembl_id	baseMean	log2FoldChange	lfcSE	pvalue	padj	gene_name
ENSMUSG00000044505	74.0480845187485	1.71797789626701	0.439744974240386	3.43750635218637e-06	0.000193326423140404	Lingo4
ENSMUSG00000007646	14.5434640260684	-1.54688192713275	0.391678403098943	3.6192781387486e-06	0.000201209677368744	Rad51c
ENSMUSG00000035745	16.5068263207717	1.52920944340881	0.394934486430846	5.06798269744679e-06	0.00025815824275548	Grin3b
ENSMUSG00000103313	40.8231287897529	2.21018692770093	0.586870220040183	7.37628169895806e-06	0.000352071866618031	Gm38357
ENSMUSG00000023078	36.4033126608795	-2.47721835141111	0.672748158828972	8.39354656501186e-06	0.000387867390631599	Cxcl13
ENSMUSG00000041293	342.851910878944	-2.03996031393003	0.553788962719193	8.39021053015753e-06	0.000387867390631599	Adgrfl
ENSMUSG00000049588	15.7664311672807	-1.59648912607105	0.445196468458205	1.47524492907155e-05	0.000614347689265848	Ccdc69
ENSMUSG00000090622	186.968621810295	2.07931874974341	0.596992867515225	1.76136912768898e-05	0.000711907132110504	A930033H14Rik
ENSMUSG00000091285	12.5426764629515	1.66480927533007	0.460608968350895	1.7926272086661e-05	0.000722528355492918	Gm17430
ENSMUSG00000098235	9.0423526380464	1.89263875018585	0.529982563592486	1.91858276480984e-05	0.000764797690038209	NA
ENSMUSG00000044927	18.6701815274399	-1.66092011588795	0.473372285624801	2.10637769487617e-05	0.000830257841100728	H1fx
ENSMUSG00000091329	14.9238965550706	1.6170186674955	0.455899291508546	2.24152480716831e-05	0.000871971178338127	1700112D23Rik
ENSMUSG00000025004	135.326894605277	-2.33292120167468	0.693578718681911	2.58684703726434e-05	0.000982595563107475	Cyp2c40
ENSMUSG00000072214	18.0952564086128	-1.63249852676583	0.478718351284391	2.93136512727684e-05	0.0010713881107503	Sept5
ENSMUSG00000084923	23.8451045529001	-1.76332092933545	0.525454064875518	3.13448285331093e-05	0.00112023020200841	Gm15611
ENSMUSG00000021453	540.427360807802	-2.39570363720549	0.729984863931386	3.28647651581548e-05	0.0011602621470677	Gadd45g
ENSMUSG00000029683	26.9885474287585	1.77962755758898	0.541412453953807	3.34303765700773e-05	0.00117451516714727	Lmod2
ENSMUSG00000086443	12.0909751392291	1.85071654481559	0.549115921662533	3.37671935891849e-05	0.00118348304101225	4933421A08Rik
ENSMUSG00000015090	95.3717341781163	-1.52962101948474	0.479170558604883	4.97657883354906e-05	0.00160919470135149	Ptgds
ENSMUSG00000006462	13.4890520143902	1.58108973392906	0.49717225944575	5.54023259477142e-05	0.00172879085914265	A530013C23Rik
ENSMUSG00000038754	6084.34344550565	-2.15476809845514	0.701577767390768	6.6537436402692e-05	0.00199063546845992	Elov13
ENSMUSG00000022096	22.2632109520546	-1.77540049077713	0.582188876729559	7.91064368348631e-05	0.00227744920332116	Hr
ENSMUSG00000029752	212.574567726209	-1.88285840191387	0.669351737134485	0.000152233894218673	0.00380682251413293	Asns
ENSMUSG00000033998	15.7718075458625	-1.74070476260931	0.606968526647088	0.000152459270445777	0.00380682251413293	Kcnk1
ENSMUSG00000087254	8.09623464906621	1.71663945665183	0.602968728803784	0.000191675630750793	0.00454446634345425	NA
ENSMUSG00000058794	26.4527269794215	-1.54676842649068	0.572930281769438	0.000240890502064942	0.00541071390861038	Nfe2

Table S2. Full significant DEG list, LV (n=20, padj<0.01, |log2FC|>1.5)

ensembl_id	baseMean	log2FoldChange	lfcSE	pvalue	padj	gene_name
ENSMUSG00000022041	145.660582378684	2.57638439138763	0.175843685950858	6.1318175414205e-50	9.18300995003134e-46	Chrna2
ENSMUSG00000027875	1200.94711694295	2.93313922478898	0.255403028018164	7.28163702360386e-32	3.63499320218305e-28	Hmgcs2
ENSMUSG00000038801	120.14262231586	-1.76007778391204	0.197567104717418	2.21194253633446e-20	3.68067238046055e-17	Scgb1c1
ENSMUSG00000020734	29.5517812480675	-1.87895225500635	0.302122564522168	2.83651105537677e-11	1.21370255900921e-08	Grin2c
ENSMUSG00000095186	19.8826870421646	3.03089625163073	0.502555274565173	5.29227988373781e-11	1.98142958847144e-08	Gm10718
ENSMUSG00000091028	31.1354029618796	2.64570833732513	0.439222559838314	8.39734026689665e-11	3.04987400456305e-08	Gm10722
ENSMUSG00000032315	10.8139329400401	-6.02992297872652	1.30126997553086	4.4898497679556e-10	1.20071410937327e-07	Cyp11a1
ENSMUSG00000023232	24.0675346558095	1.64299469724139	0.289940313243567	7.4787126441852e-10	1.89832543320877e-07	Serinc2
ENSMUSG00000048620	22.3566691030969	2.0496171970267	0.370097871919196	1.05750764999413e-09	2.62398780297812e-07	Olf1336
ENSMUSG00000075014	2251.45787365954	3.06552801252231	0.557506255662567	1.59928976755004e-09	3.74233805606709e-07	Gm10800
ENSMUSG00000095891	21.063915790526	2.88083802519752	0.526001024305662	2.38810538624774e-09	5.03722060062621e-07	Gm10717
ENSMUSG00000095280	217.188661699615	2.8743840603429	0.548218694671638	6.31642994321212e-09	1.15832159708323e-06	Gm21738
ENSMUSG00000097312	239.456629550122	3.08767833419825	0.604875992305882	1.31846837492715e-08	2.2959746956871e-06	Gm26870
ENSMUSG00000049723	21.4361424790104	2.06072610110078	0.420170005494377	4.79570759286431e-08	7.04122714811136e-06	Mmp12
ENSMUSG00000075015	207.91191270844	3.01379288346187	0.62766517825187	6.03228232122677e-08	8.4429401909058e-06	Gm10801
ENSMUSG00000025013	27.9371153821631	1.58150802858036	0.358002237498504	4.13730908907278e-07	4.2731269598589e-05	Tll2
ENSMUSG00000020051	25.730795060773	-1.95357435059826	0.458852987463658	7.5656679899915e-07	7.12600275585615e-05	Pah
ENSMUSG00000026100	19.0938326318089	2.05910611163272	0.520708593907008	4.04892180786927e-06	0.000295788551193416	Mstn
ENSMUSG00000002289	313.214505276466	1.83669349577153	0.543242960858156	2.50597877149388e-05	0.0012778693073941	Angptl4
ENSMUSG00000006403	28.2065760557275	-1.51948001831172	0.469992790167038	4.52982011755165e-05	0.00202503242031204	Adamts4

Table S3. Liver GO-BP enrichment (full, padj<0.05)

ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment
GO:0033559	unsaturated fatty acid metabolic process	9/65	127/12443	0.0708661417322835	13.5659600242277
GO:0042178	xenobiotic catabolic process	6/65	38/12443	0.157894736842105	30.2259109311741
GO:0006805	xenobiotic metabolic process	8/65	104/12443	0.0769230769230769	14.7254437869822
GO:0071466	cellular response to xenobiotic stimulus	9/65	159/12443	0.0566037735849057	10.8357039187228
GO:0019373	epoxygenase P450 pathway	5/65	27/12443	0.185185185185185	35.4501424501425
GO:0042573	retinoic acid metabolic process	5/65	28/12443	0.178571428571429	34.1840659340659
GO:0016053	organic acid biosynthetic process	11/65	298/12443	0.0369127516778524	7.06623644811564
GO:0070989	oxidative demethylation	4/65	14/12443	0.285714285714286	54.6945054945055
GO:0019369	arachidonate metabolic process	6/65	68/12443	0.0882352941176471	16.8909502262443
GO:0120254	olefinic compound metabolic process	8/65	163/12443	0.049079754601227	9.39537517697027
GO:0070988	demethylation	4/65	18/12443	0.222222222222222	42.5401709401709
GO:0008202	steroid metabolic process	10/65	288/12443	0.0347222222222222	6.64690170940171
GO:0006690	icosanoid metabolic process	7/65	117/12443	0.0598290598290598	11.4531229454306
GO:0046394	carboxylic acid biosynthetic process	10/65	295/12443	0.0338983050847458	6.48917861799218
GO:0120162	positive regulation of cold-induced thermogenesis	6/65	78/12443	0.0769230769230769	14.7254437869822
GO:0009410	response to xenobiotic stimulus	10/65	348/12443	0.028735632183908	5.50088417329797
GO:0001523	retinoid metabolic process	5/65	68/12443	0.0735294117647059	14.0757918552036
GO:0016101	diterpenoid metabolic process	5/65	71/12443	0.0704225352112676	13.4810400866739
GO:0120161	regulation of cold-induced thermogenesis	6/65	121/12443	0.0495867768595041	9.49243483788938
GO:0006631	fatty acid metabolic process	10/65	406/12443	0.0246305418719212	4.71504357711254
GO:0001676	long-chain fatty acid metabolic process	6/65	126/12443	0.0476190476190476	9.11575091575092
GO:0106106	cold-induced thermogenesis	6/65	126/12443	0.0476190476190476	9.11575091575092
GO:0046456	icosanoid biosynthetic process	4/65	43/12443	0.0930232558139535	17.8075134168157
GO:0006721	terpenoid metabolic process	5/65	84/12443	0.0595238095238095	11.3946886446886
GO:0072330	monocarboxylic acid biosynthetic process	7/65	199/12443	0.0351758793969849	6.73374565133359
GO:1990845	adaptive thermogenesis	6/65	138/12443	0.0434782608695652	8.32307692307692
GO:0006636	unsaturated fatty acid biosynthetic process	4/65	48/12443	0.0833333333333333	15.9525641025641
GO:0001659	temperature homeostasis	6/65	152/12443	0.0394736842105263	7.55647773279352
GO:0006720	isoprenoid metabolic process	5/65	99/12443	0.0505050505050505	9.66822066822067
GO:0035634	response to stilbenoid	3/65	24/12443	0.125	23.9288461538462

ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment
GO:0042445	hormone metabolic process	6/65	187/12443	0.0320855614973262	6.14216371863431
GO:0008210	estrogen metabolic process	3/65	29/12443	0.103448275862069	19.8031830238727
GO:0034756	regulation of iron ion transport	2/65	10/12443	0.2	38.2861538461539
GO:0050910	detection of mechanical stimulus involved in sensory perception of sound	2/65	10/12443	0.2	38.2861538461539

Table S4. LV GO-BP enrichment (full, padj<0.05)

ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment
GO:0048630	skeletal muscle tissue growth	2/13	10/13056	0.2	200.861538461538
GO:0016114	terpenoid biosynthetic process	2/13	17/13056	0.117647058823529	118.153846153846
GO:0061450	trophoblast cell migration	2/13	21/13056	0.0952380952380952	95.6483516483516
GO:0033574	response to testosterone	2/13	28/13056	0.0714285714285714	71.7362637362637
GO:0008299	isoprenoid biosynthetic process	2/13	29/13056	0.0689655172413793	69.262599469496
GO:0006516	glycoprotein catabolic process	2/13	30/13056	0.0666666666666667	66.9538461538462
GO:0044703	multi-organism reproductive process	3/13	163/13056	0.0184049079754601	18.4841906559698
GO:0044706	multi-multicellular organism process	3/13	165/13056	0.0181818181818182	18.2601398601399
GO:1901654	response to ketone	3/13	185/13056	0.0162162162162162	16.2860706860707
GO:0044283	small molecule biosynthetic process	4/13	487/13056	0.0082135523613963	8.24893381772232

Table S5. Liver KEGG pathway enrichment (full, padj<0.05)

category	subcategory	ID	Description	GeneRatio	BgRatio
Metabolism	Lipid metabolism	mmu00140	Steroid hormone biosynthesis	10/40	69/5954
Metabolism	Metabolism of cofactors and vitamins	mmu00830	Retinol metabolism	10/40	77/5954
Human Diseases	Cancer: overview	mmu05204	Chemical carcinogenesis - DNA adducts	9/40	67/5954
Human Diseases	Cancer: overview	mmu05207	Chemical carcinogenesis - receptor activation	9/40	155/5954
Metabolism	Lipid metabolism	mmu00591	Linoleic acid metabolism	5/40	28/5954
Metabolism	Lipid metabolism	mmu00590	Arachidonic acid metabolism	6/40	53/5954
Metabolism	Xenobiotics biodegradation and metabolism	mmu00982	Drug metabolism - cytochrome P450	4/40	53/5954
Metabolism	Xenobiotics biodegradation and metabolism	mmu00980	Metabolism of xenobiotics by cytochrome P450	4/40	57/5954
Metabolism	Xenobiotics biodegradation and metabolism	mmu00983	Drug metabolism - other enzymes	4/40	82/5954

Table S6. STRING network hub genes (Angptl4-Sdc/Cdh5 axis + first-shell interactors)

gene	degree	betweenness	is_seed
Sdc1	16	64.3	TRUE
Sdc4	14	45.1	TRUE
Sdc3	6	3.2	TRUE
Gpc1	6	2.1	FALSE
Gpc4	5	1	FALSE
Hspg2	5	0.6	FALSE
Ext1	4	0.2	FALSE
Tnc	3	0.4	FALSE
Sdc2	3	0	TRUE
Angptl4	2	1	TRUE
Comp	2	0	FALSE
Sdcbp2	2	0	FALSE
Tnr	2	0	FALSE
Thbs3	2	0	FALSE
Tnxb	2	0	FALSE
Tnn	2	0	FALSE
Gpc2	2	0	FALSE
Cdh5	1	0	TRUE
Angptl8	1	0	FALSE
Nudt1611	1	0	FALSE
Hpse2	1	0	FALSE

Table S7. Full AutoDock Vina ranking (all 21 compounds)

label	best_affinity_kcal_mol	status
Resmetirom	-9.068	OK
Ezetimibe	-8.883	OK
Pioglitazone	-8.279	OK
Fenofibrate	-8.206	OK
Empagliflozin	-8.144	OK
Bezafibrate	-8.101	OK
Rosiglitazone	-7.841	OK
Ibuprofen	-7.233	OK
Dapagliflozin	-7.05	OK
Arachidonic_acid	-6.995	OK
Atorvastatin	-6.831	OK
Obeticholic_acid	-6.775	OK
Eicosapentaenoic_acid	-6.772	OK
Aspirin	-6.703	OK
Icosapent_ethyl	-6.693	OK
Metoprolol	-6.493	OK
Linoleic_acid	-6.467	OK
Palmitic_acid	-6.38	OK
Caffeine	-5.358	OK
Niacin	-5.078	OK
Metformin	-4.654	OK

Table S8. STITCH chemical partners of Angptl4-axis genes (full)

gene	cid	raw_id	compound_name
Angptl4	3931	-10.CID100003931	9,12-Octadecadienoic Acid
Angptl4	77998	-10.CID100077998	C22H23NO7S
Angptl4	985	-10.CID100000985	Palmitic Acid
Angptl4	231	-10.CID100000231	5,8,11,14-Eicosatetraenoic acid, (5Z,8Z,11Z,14Z)-
Angptl4	860	-10.CID100000860	Linseed Acid
Angptl4	1424	-10.CID100001424	13-Hydroxyoctadeca-9,11-dienoic acid
Angptl4	1927	-10.CID100001927	9-Hydroxy-10,12-Octadecadienoic Acid
Angptl4	3209	-10.CID100003209	5,8,11,14,17-Eicosapentaenoic Acid
Angptl4	4829	-10.CID100004829	Pioglitazone
Sdc1	18845	-10.CID100018845	Iduronic Acid
Sdc1	150835	-10.CID100150835	[3,7-Dimethyl-9-(2,6,6-trimethylcyclohexen-1-yl)nona-2,4,6,8-tetraenyl] octadeca-9,12-dienoate
Sdc1	122008	-10.CID100122008	9-cis-RetinylStearate-d5
Sdc1	961	-10.CID100000961	Hydroxide
Sdc1	175	-10.CID100000175	Acetate
Sdc1	1172	-10.CID100001172	(5-(2,4-Dioxypyrimidin-1-yl)-3,4-dihydroxyoxolan-2-yl)methyl dihydrogen phosphate
Sdc3	18845	-10.CID100018845	Iduronic Acid
Sdc3	175	-10.CID100000175	Acetate
Sdc2	18845	-10.CID100018845	Iduronic Acid
Cdh5	271	-10.CID100000271	Calcium Cation

Table S9. DGIdb druggability summary

gene	n_interactions	n_approved	example_drugs
SDC2	1	0	HEPARAN SULFATE
SDC4	1	1	REPOTRECTINIB
CDH5	1	0	FX06
SDC3	0	0	
SDC1	3	0	HEPARIN; INDATUXIMAB RAVTANSINE
ANGPTL4	0	0	

Supplemental Text

Text S1. Final target selection rationale

Why not the syndecans (the actual network hubs): all available PDB structures for SDC1/SDC4 cover fragments other than the extracellular heparan-sulfate-bearing ectodomain that would physically engage a secreted ligand like Angptl4 (e.g. 8BLV is the SDC4 cytoplasmic tail bound to syntenin PDZ domains, not an extracellular pocket). Docking a small molecule against a cytoplasmic PDZ-binding motif would not address the actual Angptl4-syndecan extracellular interaction, so despite their strong network centrality, they are not structurally actionable for this step.

Why ANGPTL4: (1) zero existing DGIdb drug interactions -- genuinely undrugged; (2) a real, solved, 2.3-angstrom crystal structure of its C-terminal fibrinogen-like domain exists (6EUB) with a visible ligand-accommodating surface (PEG fragment bound in the deposited structure); (3) it is the molecule directly validated twice over -- by our own stricter DEG reanalysis (LV upregulated, padj/lfc-significant) and by an independent 2024 single-cell paper confirming cardiac-fibroblast secretion in HFpEF; (4) blocking this domain would act at the source of the axis (the secreted signal itself) rather than downstream, which is defensible as the more tractable intervention point given the receptor-side structures are unusable.

Limitation to state plainly in the paper: this reframes the intervention target from the network-centrality hub (Sdc1/Sdc4) to the ligand itself, because of a structural-biology constraint, not a change in the network analysis's conclusions -- Sdc1/Sdc4 remain the top computational hubs, but are not currently actionable for small-molecule structure-based screening with public structural data.

Text S2. DiffDock cross-check summary

Vina docking (Section above) used a fixed search box centered on the only observed small-molecule-occupied surface site in the ANGPTL4 crystal structure (the 1PE/PEG cryoprotectant site). DiffDock, by contrast, performs blind docking across the entire protein surface with no box constraint — so agreement between the two methods on **where** a ligand binds is an independent structural check, not just a re-ranking of the same fixed site.

- Ezetimibe is the more cross-validated candidate: an orthogonal, box-free method (DiffDock) independently converged on essentially the same surface pocket that Vina was constrained to search, and did so with a **better** confidence score than Resmetirom. Contact residues at this site (Leu304, Leu312, Gly313, Ala314, Leu322, Ser323, Val324, Trp349, Trp350, Gly352, Thr353, His356) form a real hydrophobic groove consistent with a genuine small-molecule pocket, not a surface artifact.
- Resmetirom scored best on Vina (which only searched the predefined pocket), but DiffDock's unconstrained search preferred a completely different surface region (contact residues Trp280, Asp281, Ile367, Leu374, Tyr387, Tyr388, Pro389). This is reported as a genuine discrepancy rather than smoothed over: Resmetirom's top Vina score should be read as "the best-fitting pose **within the pocket we searched**," not as strong independent evidence that this is its preferred binding site on ANGPTL4.
- Both DiffDock confidence scores (-2.06, -1.32) fall in DiffDock's own low-to-moderate confidence range (published benchmarks treat $c > 0$ as high-confidence), so neither result should be oversold as a confirmed binder -- both are hypothesis-generating leads for this 학술제 pipeline, not validated hits.

DiffDock was not run locally: this machine has no CUDA-capable GPU, and a local from-scratch install (PyTorch + PyTorch Geometric + e3nn + ESM) would have made CPU inference impractically slow for even one complex. Instead, a publicly running DiffDock Gradio Space (swcanner/DiffDock-Web, mirroring the original gcorso/DiffDock inference code) was used via its web UI for the two approved top-hit compounds only, consistent with the earlier CPU-budget decision to limit DiffDock to the final 1-2 candidates rather than the full 21-compound library.